

V3 empirical study · Instruction

Five models · four underlyings · four volatility regimes · 1.5-year monthly calibration · return-based calibration

What this report is

- Companion to the seven-model 10,000-path study. This report keeps GBM, Modified GBM, GARCH, Merton, and GARCH-Merton, which are calibrated from lookback returns. Rankings are within this group; they are not compared here with option-implied Heston models.
- Companies are SPY, AAPL, MSFT, and AMZN. No new calibration or LSM. Every number is the stored cell from V3_1p5y_monthly_empirical_study.
- Option-implied companion: V3_1p5y_monthly_empirical_study_option_implied.pdf.
- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 5 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: GBM, Modified GBM (Markov direction), GARCH(1,1), Merton jump-diffusion, GARCH-Merton.
- Modified GBM (Markov direction + split-normal magnitudes) is specified in V3_modified_gbm_model.pdf in this folder.
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Five models · four underlyings · four volatility regimes · 1.5-year monthly calibration · return-based calibration

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all five models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean} |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all five models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with $\text{trading_date} \leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All five models in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	55.01	1.4054	+0.2705	+21.16	24.5%	5
Modified GBM	47.67	1.2488	-0.0854	+10.62	25.9%	1
GARCH	52.86	1.6883	+1.6456	+43.57	20.8%	2
Merton	52.87	1.3685	+0.1826	+18.56	20.3%	3
GARCH-Merton	54.76	1.7780	+1.7384	+46.28	20.5%	4

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	38.08	0.6850	+0.4678	+24.88	26.1%	3
Modified GBM	37.08	0.6750	+0.4371	+23.66	26.6%	1
GARCH	38.24	0.6860	+0.4700	+24.96	23.3%	4
Merton	37.52	0.6728	+0.4520	+24.20	24.7%	2
GARCH-Merton	42.83	0.7744	+0.6153	+30.82	22.5%	5

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	49.51	1.6628	+1.1528	+32.90	21.8%	2
Modified GBM	51.83	1.7487	+1.2257	+34.67	22.2%	3
GARCH	71.72	2.1416	+1.7421	+45.61	21.9%	4
Merton	47.62	1.5845	+1.0524	+30.44	16.2%	1
GARCH-Merton	77.02	2.4122	+2.1360	+54.80	20.5%	5

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	64.88	3.6003	+1.6141	+44.13	23.9%	4
Modified GBM	64.56	3.4810	+1.5419	+44.09	24.2%	3
GARCH	56.59	2.9250	+2.0268	+34.97	32.7%	1
Merton	60.38	3.3441	+1.2734	+38.99	18.7%	2
GARCH-Merton	66.14	3.4255	+2.8053	+46.06	31.4%	5

Page 5 · Table block · Company summary · five models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green row = lowest mean RMSE%. 1.5-year lookback, rolling=monthly, n_paths=

Table Co-1 Company summary · 2008-2009 Crisis

2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	49.54	1.4392	+0.7383	+23.71	23.1%	3
Modified GBM	44.73	1.2794	+0.4468	+17.76	24.6%	1
GARCH	52.30	1.5524	+1.3564	+39.86	21.3%	4
Merton	48.36	1.4136	+0.6645	+21.87	19.1%	2
GARCH-Merton	62.67	1.8758	+1.7623	+51.27	20.1%	5

Table Co-2 Company summary · 2013-2014 Normal

2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	47.93	2.2358	+1.7353	+36.05	24.6%	3
Modified GBM	44.77	2.0621	+1.5246	+33.19	25.4%	2
GARCH	58.97	2.3886	+2.0506	+45.16	22.8%	4
Merton	42.53	2.0512	+1.4116	+30.28	18.4%	1
GARCH-Merton	79.38	3.5675	+3.5143	+67.98	18.9%	5

Table Co-3 Company summary · 2018-2019 Late-cycle

2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n} = 51$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	39.31	8.0415	+4.4076	+23.84	23.2%	3
Modified GBM	38.45	8.4064	+5.2944	+22.84	24.1%	2
GARCH	43.38	7.7461	+5.7535	+28.81	19.5%	4
Merton	37.91	8.0116	+3.8942	+21.67	19.3%	1
GARCH-Merton	52.78	9.0776	+7.7086	+39.80	17.7%	5

Table Co-4 Company summary · 2019-2020 COVID

2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	48.18	10.6159	+4.7227	+30.10	24.0%	3
Modified GBM	47.07	10.6143	+4.7108	+28.16	24.6%	2
GARCH	52.33	9.6856	+5.4722	+33.95	24.5%	4
Merton	46.08	10.3990	+4.1300	+27.28	19.4%	1
GARCH-Merton	153.02	12.2225	+9.1243	+58.27	23.5%	5

RMSE%, MAE, Bias \$, Bias%, and early-exercise fraction are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-pricing of a combined contract sample. Ranking uses the mean RMS

Page 6 · Table block · AAPL · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	57.67	2.4326	+1.3849	+31.10	22.8%	4
Modified GBM	51.94	2.1591	+0.9309	+23.67	23.5%	2
GARCH	48.71	2.7044	+2.5500	+40.72	19.1%	1
Merton	55.67	2.3640	+1.2255	+28.42	18.3%	3
GARCH-Merton	58.94	3.2224	+3.1062	+50.39	15.9%	5

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	50.20	3.5766	+3.5387	+38.63	24.6%	4
Modified GBM	45.44	3.1559	+3.1131	+34.56	25.4%	2
GARCH	45.44	3.1849	+3.0454	+29.67	22.2%	1
Merton	45.50	3.2302	+3.1799	+33.94	19.5%	3
GARCH-Merton	65.87	4.9053	+4.9027	+53.60	17.8%	5

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	32.53	1.7033	+1.0081	+18.76	23.9%	2
Modified GBM	34.33	1.7907	+1.1260	+20.69	24.0%	3
GARCH	48.45	2.0164	+1.5757	+29.40	20.0%	4
Merton	30.59	1.5914	+0.8543	+16.31	19.9%	1
GARCH-Merton	59.31	2.6521	+2.3998	+42.76	18.5%	5

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	46.19	3.9995	+2.3663	+30.76	23.8%	3
Modified GBM	45.88	3.7712	+2.1029	+29.60	24.4%	2
GARCH	61.17	4.8546	+4.4651	+43.67	24.2%	4
Merton	43.61	3.6886	+1.9753	+27.36	18.9%	1
GARCH-Merton	71.81	5.9515	+5.7532	+57.54	24.2%	5

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 7 · Table block · MSFT · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	44.02	0.3723	+0.0897	+17.99	24.1%	3
Modified GBM	41.99	0.3590	+0.0799	+16.83	25.2%	1
GARCH	70.79	0.5879	+0.5440	+54.27	30.0%	4
Merton	42.94	0.3684	+0.0716	+16.18	19.7%	2
GARCH-Merton	81.24	0.6935	+0.6704	+66.90	29.6%	5

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	62.49	0.4283	+0.4261	+52.98	24.7%	3
Modified GBM	59.39	0.4056	+0.4025	+50.16	25.7%	2
GARCH	90.24	0.6511	+0.6511	+78.36	25.2%	4
Merton	54.85	0.3679	+0.3602	+45.39	17.1%	1
GARCH-Merton	112.93	0.8656	+0.8656	+102.63	21.6%	5

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	41.73	1.0839	+0.8054	+27.55	23.0%	3
Modified GBM	33.44	0.8877	+0.4849	+17.78	25.3%	1
GARCH	42.43	1.0703	+0.9006	+30.13	18.9%	4
Merton	40.33	1.0504	+0.7474	+25.77	19.8%	2
GARCH-Merton	52.89	1.3664	+1.2944	+42.08	17.0%	5

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	45.93	2.0118	+1.0018	+28.12	24.5%	3
Modified GBM	40.97	1.7579	+0.6597	+22.12	25.1%	1
GARCH	60.29	2.3979	+1.9268	+35.85	29.6%	4
Merton	44.13	1.8737	+0.7910	+25.02	19.7%	2
GARCH-Merton	346.80	6.7038	+6.3160	+88.26	28.7%	5

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 8 · Table block · AMZN · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	46.92	1.5126	+0.7403	+22.05	22.4%	4
Modified GBM	40.25	1.3199	+0.3297	+12.79	24.9%	2
GARCH	37.41	1.3649	+0.9751	+24.58	14.6%	1
Merton	46.47	1.5086	+0.6964	+21.02	19.3%	3
GARCH-Merton	47.82	1.7115	+1.5102	+36.53	14.8%	5

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	31.09	2.7024	+1.2412	+16.54	24.6%	3
Modified GBM	29.48	2.6249	+1.0584	+14.84	25.1%	2
GARCH	41.25	3.3299	+2.4552	+27.44	21.0%	4
Merton	27.23	2.5555	+0.6946	+11.52	18.7%	1
GARCH-Merton	59.35	4.9315	+4.7746	+47.73	17.4%	5

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	43.66	21.3373	+11.4095	+25.21	22.9%	3
Modified GBM	47.59	22.5407	+14.2723	+30.06	23.0%	5
GARCH	39.26	20.1516	+14.7843	+26.92	19.6%	1
Merton	42.81	21.3929	+10.0809	+22.92	18.1%	2
GARCH-Merton	46.13	23.2142	+19.4315	+34.56	17.6%	4

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	52.42	25.8364	+10.7999	+31.41	23.8%	4
Modified GBM	54.35	26.3138	+11.3697	+32.76	24.2%	5
GARCH	35.53	21.8043	+10.0246	+22.34	19.6%	1
Merton	50.50	25.6348	+9.6238	+29.46	19.7%	3
GARCH-Merton	40.45	24.0122	+15.3037	+29.03	17.5%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Merton	45.19	44.82	5
2	Modified GBM	45.39	45.66	5
3	GBM	47.65	46.56	0
4	GARCH	52.52	48.58	6
5	GARCH-Merton	80.27	59.33	0

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
GBM	0	0	0	0	0
Modified GBM	2	0	3	0	5
GARCH	1	2	0	3	6
Merton	1	2	1	1	5
GARCH-Merton	0	0	0	0	0

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	50.90	45.47	41.86	52.35	47.65
Modified GBM	45.46	42.85	41.80	51.44	45.39
GARCH	52.44	53.79	50.47	53.40	52.52
Merton	49.49	41.28	40.34	49.66	45.19
GARCH-Merton	60.69	70.24	58.84	131.30	80.27

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	51.87	52.26	0
Modified GBM	50.29	49.75	2
GARCH	54.85	54.73	1
Merton	49.60	50.24	1
GARCH-Merton	60.19	60.45	0

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	46.65	48.20	0
Modified GBM	44.40	45.66	0
GARCH	50.94	48.58	2
Merton	43.84	44.56	2
GARCH-Merton	63.98	62.59	0

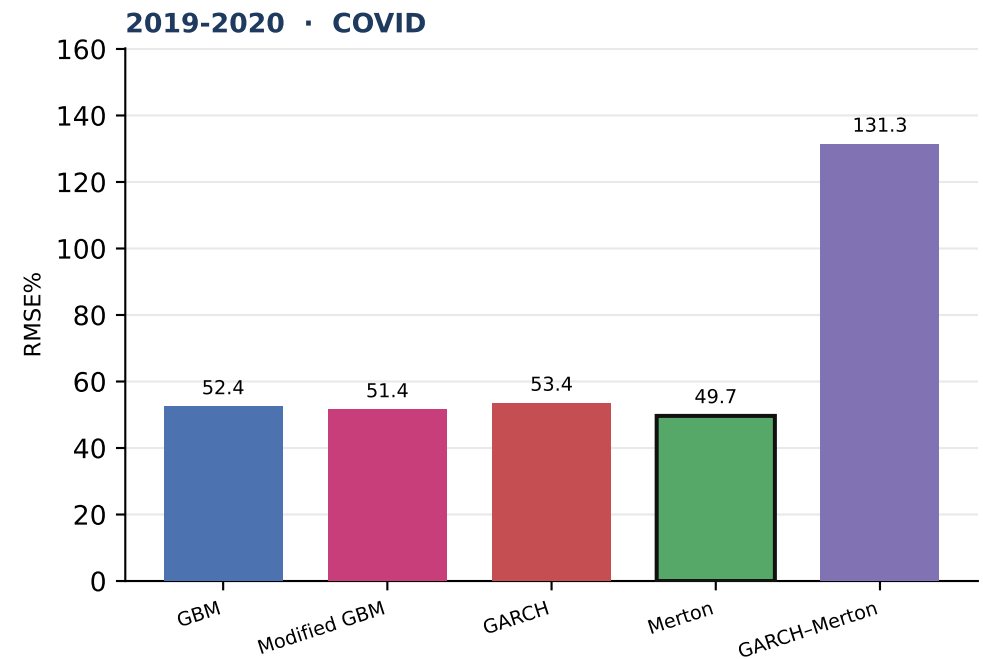
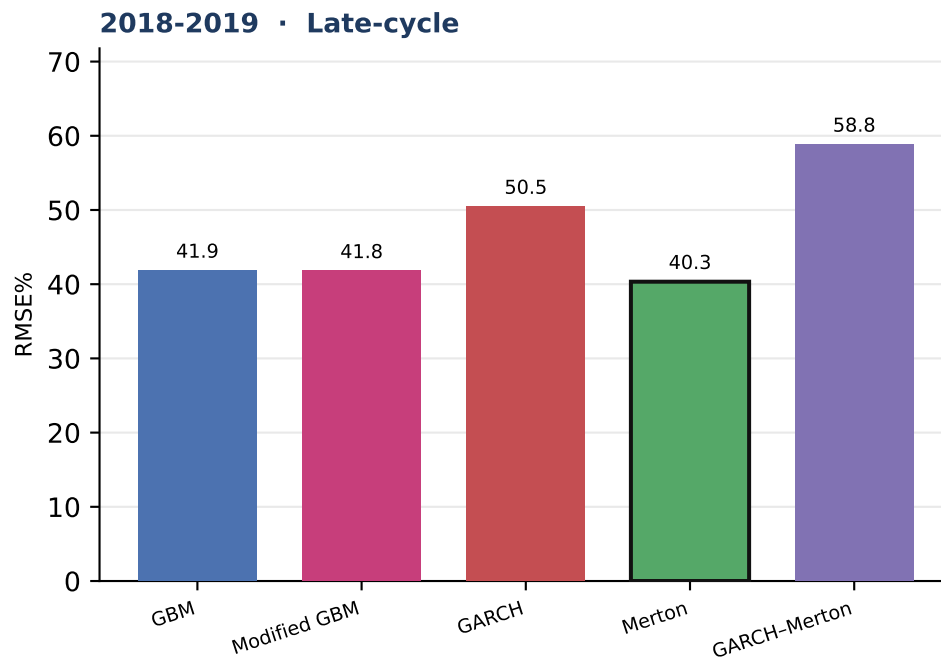
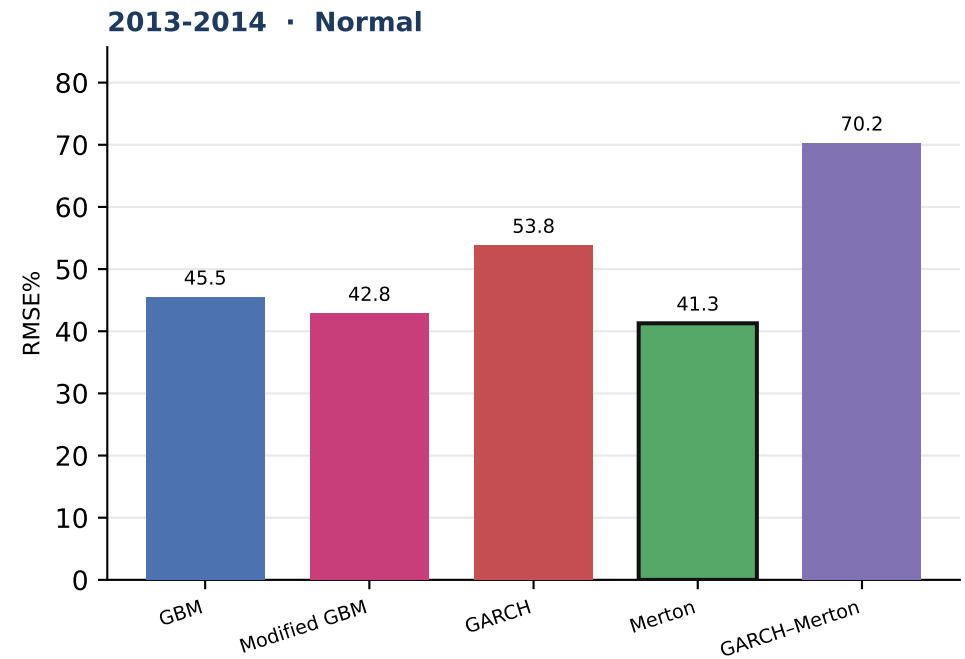
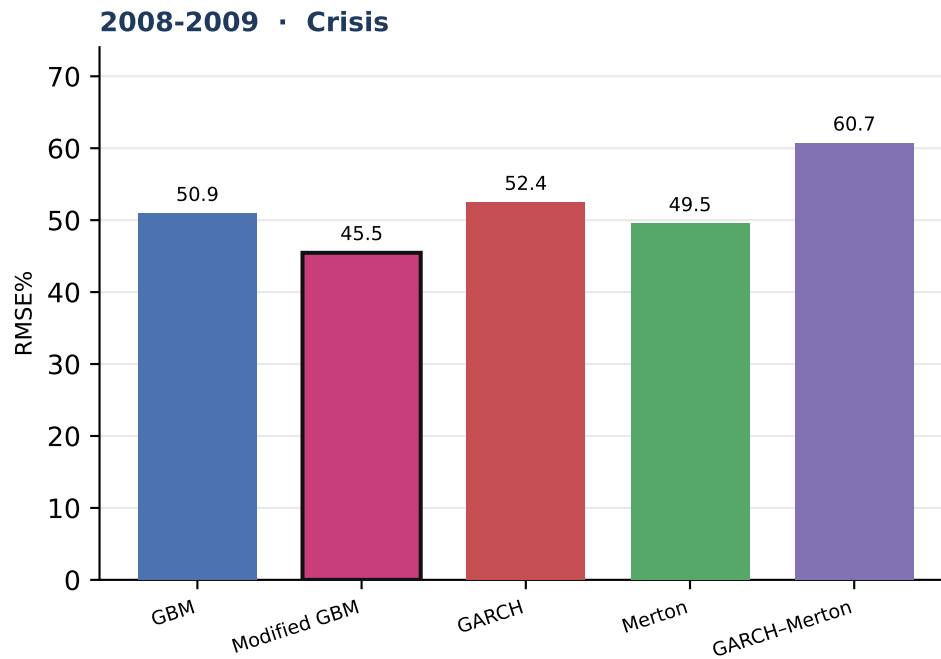
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
GBM	48.54	44.98	0
Modified GBM	43.95	41.48	3
GARCH	65.94	65.54	0
Merton	45.56	43.54	1
GARCH-Merton	148.47	97.09	0

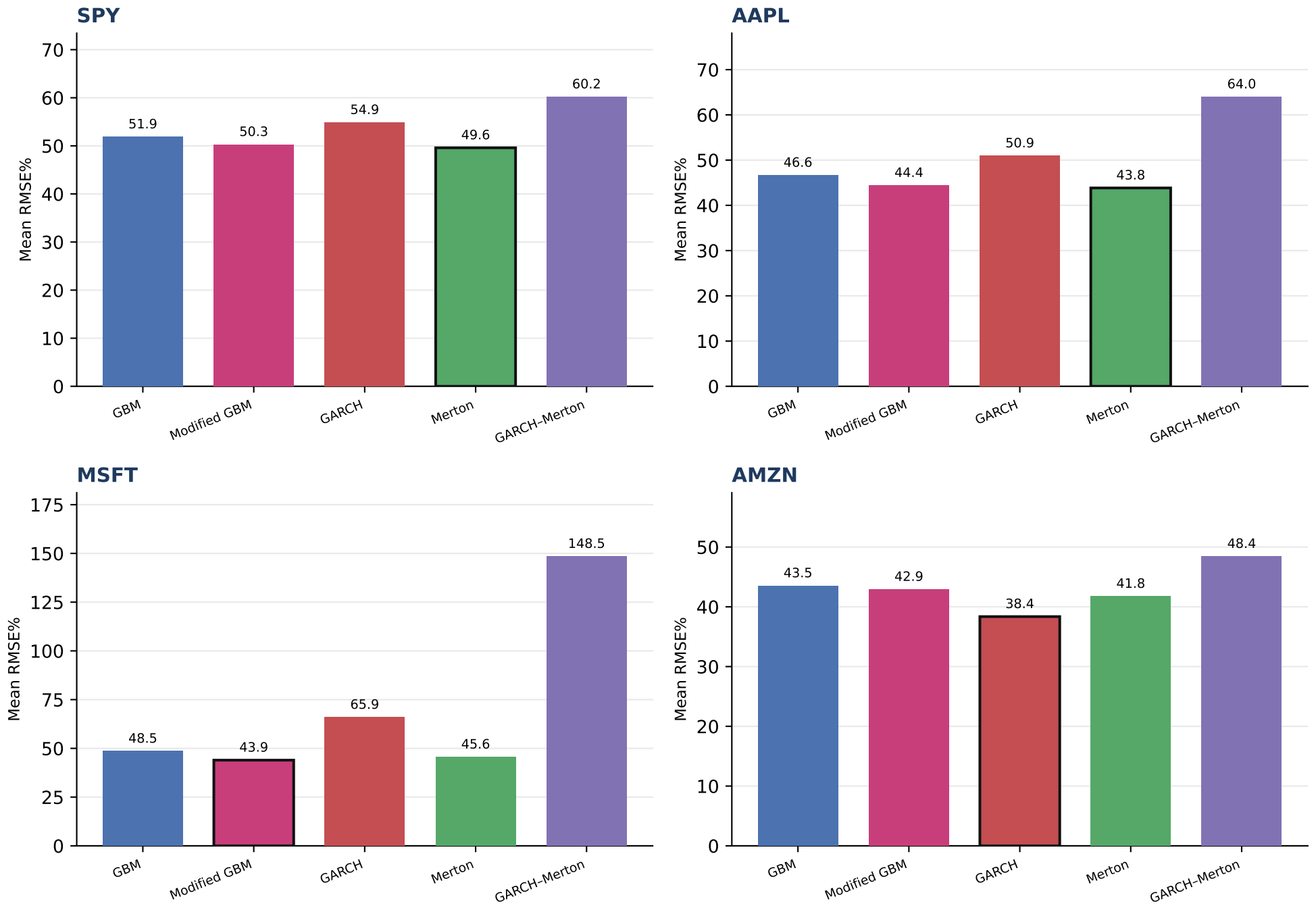
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
GBM	43.52	45.29	0
Modified GBM	42.92	43.92	0
GARCH	38.36	38.33	3
Merton	41.75	44.64	1
GARCH-Merton	48.44	46.98	0

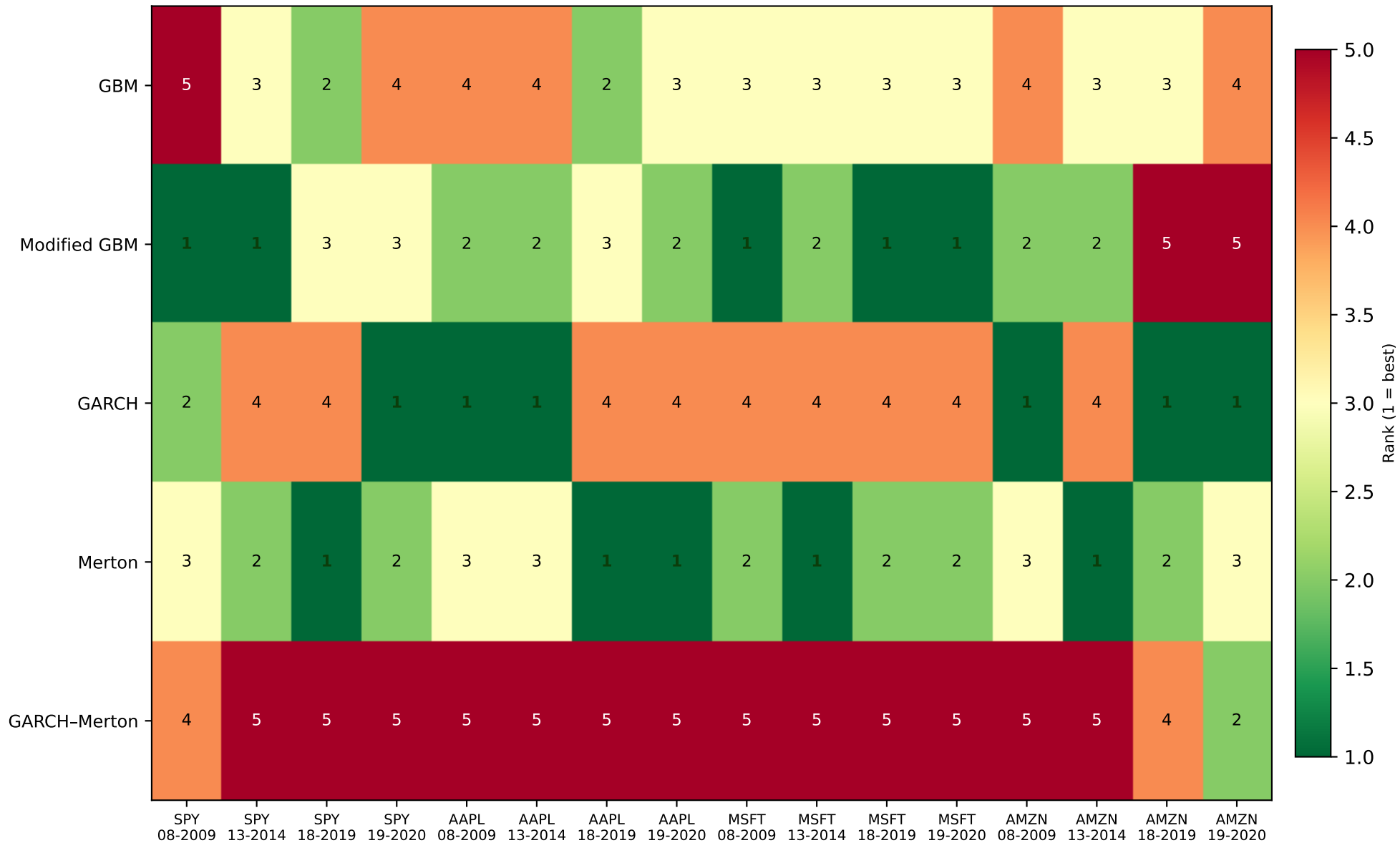
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



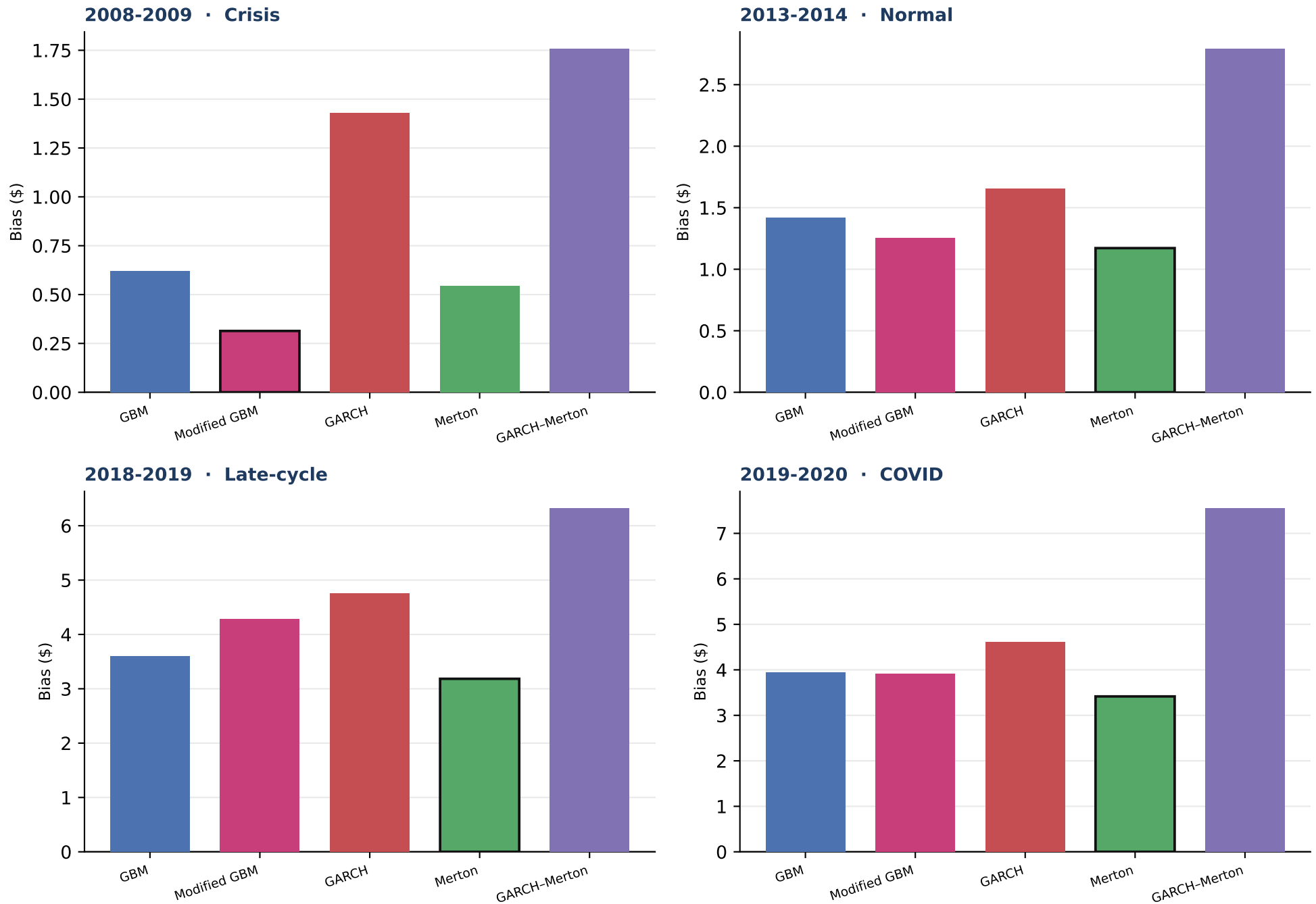
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, Merton is the best overall model (mean RMSE% = 45.19; 5 of 16 cells).
- Second overall is Modified GBM (mean RMSE% = 45.39; 5 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: GARCH, Merton, Modified GBM.
- SPY: lowest mean RMSE% is Merton (49.60); wins 1 of 4 regimes.
- AAPL: lowest mean RMSE% is Merton (43.84); wins 2 of 4 regimes.
- MSFT: lowest mean RMSE% is Modified GBM (43.95); wins 3 of 4 regimes.
- AMZN: lowest mean RMSE% is GARCH (38.36); wins 3 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Modified GBM (45.46).
- 2013-2014 (Normal): best mean RMSE% is Merton (41.28).
- 2018-2019 (Late-cycle): best mean RMSE% is Merton (40.34).
- 2019-2020 (COVID): best mean RMSE% is Merton (49.66).
- No-jump models as a group have lower mean RMSE% (48.52) than the jump models in this report (62.73).
- For the overall winner (Merton), crisis RMSE% is 49.49 vs 41.28 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all five models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).

Early-exercise fractions are a model implication, not an accuracy score. American calls on these tenors often have modest early-exercise value.

Notebook: [Early-exercise-fractions](#) is an interactive notebook with 10,000 paths at 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. [ipynb](#)
Engine: V3-Models_result/scripts/run_v3_1p5y_10k_monthly_empirical_study.py · payload.json is the shared number store.